

Dan North

PGCE MSc



22, Midland Place,
Swansea
SA7 9QU



+44 7447 459269



northdan51@gmail.com



github.com/dannorth42



dannorth42.github.io/
My_Portfolio

About Me

Having attained both my physics degrees and PGCE, I intend to continue my career within the education sector by working with scientific outreach programmes. My love for science led me into a career where I could use that to entertain, encourage and educate young people. I'm drawn to outreach as I think it's incredibly important to engage with young people and support those with an interest in STEM fields. Engaging the public is incredibly important and as ambassadors to science we should strive to be open and informative.

Areas of Specialization

Education • Physics
• Computer Science

Interests

Science Content
Sci-Fi
Gaming
LEGO
Archery
Rock Climbing
Escape Rooms

RESUMÉ

2022–

Delivery Officer

TECHNOCAMPS · Swansea University

Technocamps is a pan-Wales outreach project which aims to support and improve digital literacy across Wales. This role has given me countless opportunities to improve my teaching methods, development skills, programming ability and much more. Responsibilities include:

- Deliver computer science workshops bilingually in schools (ages 3 – 18), teacher training events and conferences, adult sessions and micro-credentials
- Represent the project at events such as Robotics Competition, Conferences, Science Festivals, Eisteddfod, Summer Camps
- Developing new content for all of the above
- Keep up to date with Welsh education including the Curriculum for Wales, DCF and WJEC curriculums
- Translate all resources into Welsh
- Collaborate with other projects and organisations such as the micro:bit Foundation, Hwb, Amgueddfa Cymru, Theatr na nOg and more



2020–2025

Manager

THE ESCAPE GAME · Swansea

I thoroughly enjoyed this part-time role, and I found watching peoples thought-process in solving puzzles fascinating. Responsibilities included:

- Leading a team of 10 game masters, including producing weekly rotas, acting as HR for staff, and running escape rooms
- Communicating with customers
- Creating and posting social media content
- Monitoring and purchasing supplies
- Upkeep and maintenance of the rooms
- Develop and maintain the website
- Design and build new escape rooms



2018–2020

Instructor & Manager

MAD SCIENCE · South Wales

At Mad Science South Wales I began my career in STEM Outreach. This gave me the opportunity to develop my teaching practices over a much larger age range while visiting a wide variety of schools across South Wales. Responsibilities included:

- Delivering engaging workshops, assemblies, after-school clubs and summer camps
- Develop new workshops and experiments
- Prepare activity kits for all staff
- Manage all equipment (both instruments and consumables), ensure that everything was in working order (and safe for use in schools)
- Manage all chemicals, including correct storage and labelling, monitoring expiration date, diluting and combining reagents
- Monitoring stock levels and regularly submit purchase requests



2016–2020

Physics Teacher

YGG BRYN Tawe & NUMEROUS AGENCIES · South Wales

Upon graduating with a PGCE I immediately secured a role in a Welsh Medium school as a Physics Teacher. After a little over a year I progressed to long-term supply with numerous agencies. Responsibilities included:

- Teaching classes from Year 7 – 13
- Redeveloping the Physics curriculum for GCSE and A-Level
- Preparing the Progression Step 3 curriculum for then the upcoming Curriculum for Wales
- Marking classwork and exams of learners and providing meaningful feedback
- Conducting parents evenings
- Liaising with other teachers across the school
- Creating and delivering ABCh (Personal and Social Education) lessons



Languages



English
Welsh

Programming



Python
Java
L^AT_EX
HTML/CSS
PHP
JavaScript

EDUCATION

- 2023– **Applied Software Engineering**
BSc · Swansea University 
Currently completing a full time Degree Apprenticeship funded by the Welsh Government.
Dissertation is an investigation into teaching the mechanisms of DNA and inheritance through their Computer Science equivalents — with learners building a working model in Python.
- 2022 **Medical Radiation Physics**
MSc · Swansea University 
Merit Awarded
First Class Dissertation: "A report evaluating the risk of unintended exposures at Swansea Bay University Health Board using voluntarily reported radiotherapy errors"
- 2016 **Secondary Physics (Welsh Medium)**
PGCE · University of Wales Trinity St. David 
Awarded with Qualified Teacher Status
Placements in Ysgol Y Strade & Ysgol Gyfun Ystalyfera
Dissertation: "An Investigation into Utilising 'Inference Squares' in Tutoring Pupils to Use the Techniques Required in Answering PISA Questions"
- 2015 **Physics**
BSc · Nottingham Trent University 
2.2. with Honours
First Class Dissertation: "A Study Into the Accountability of Evaporation in the Mass Reduction of the Aral Sea"



CERTIFICATES & QUALIFICATIONS

- 2025 Adobe Cymru Create
2024 Show Racism The Red Card — Anti-Racist Training
2024 Ambassador for CyberFirst Wales
2023 NDEC CyberFirst Deliverer
2023 First Aid at Work Certification
2023 Python Advanced micro-credential
2022 Python micro-credential
2022 Minecraft Education Qualified Deliverer
2022 Safeguarding in Schools Training
2013 Full UK Driving Licence